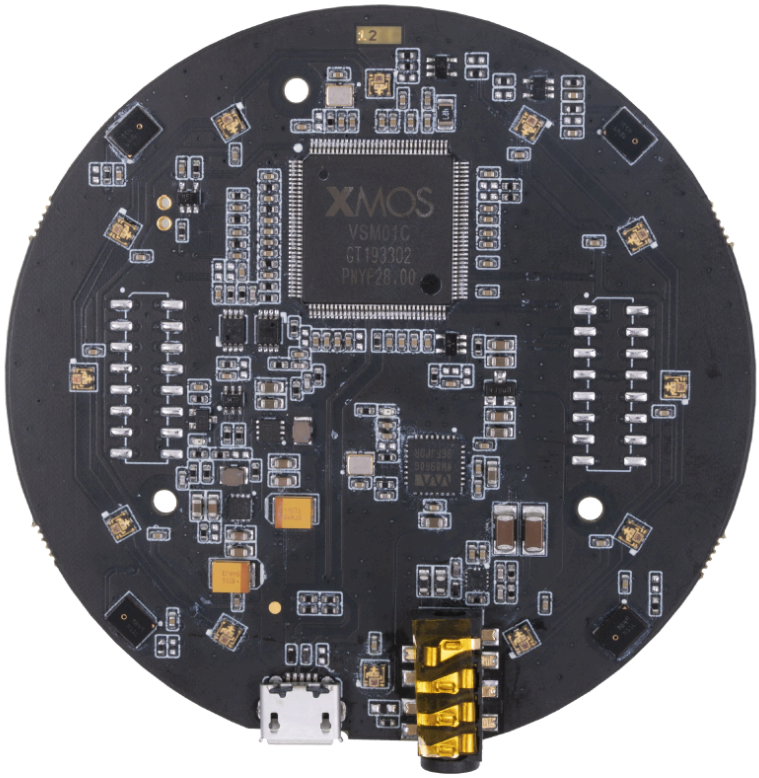
reSpeaker USB 4-Mic Array XVF3000 v3.0



We are excited to formally introduce the **reSpeaker XVF3800** — a comprehensive upgrade to the reSpeaker XVF 3000. Building upon its predecessor's foundation of 4-microphone array architecture, universal compatibility (Windows / macOS / Linux), and dual-firmware (I2S / USB) plug-and-play convenience, the XVF3800 delivers a significant leap in both **audio fidelity and algorithmic performance**.

Key Upgrade Highlights

- **AI-Powered Audio Algorithms**: Integrated suite featuring AEC (Acoustic Echo Cancellation), AGC (Automatic Gain Control), DoA (Direction of Arrival) detection, beamforming, VAD (Voice Activity Detection), noise suppression, and dereverberation — laying robust groundwork for advanced voice applications.
- **360° Far-Field Voice Capture**: Precision voice pickup within a 5-meter radius, effortlessly accommodating conferencing systems, intelligent interaction, and voice-controlled scenarios.

- **Dual Operating Modes**: Flexible USB/I2S firmware switching to meet diverse development and deployment requirements.
- **Product Details & Specifications**: ReSpeaker XVF3800 4-Mic Array Store Page
- **Quick Start & Wiki Guide**: reSpeaker XVF3800 Getting Started Guide | Seeed Studio Wiki

ReSpeaker Mic Array v3.0 is the next evolution of Seeed Studio's USB microphone arrays, following the ReSpeaker Mic Array v2.0. While the v2.0 was built on XMOS's XVF-3000 chipset and designed as a major upgrade from v1.0, the v3.0 focuses on refining audio quality and algorithm performance, even with a smaller physical microphone count.

Compared with the v2.0's 4-mic array, the v3.0 also uses 4 microphones but integrates improved built-in audio processing algorithms, offering clearer far-field voice capture and better noise handling than its predecessor. The v3.0 replaces the WM8960 codec in v2.0 with a TLV320AIC3104 codec, contributing to higher fidelity sound capture.

While the v2.0 was often paired with the ReSpeaker Core or used as a development board, the v3.0 is more of a plug-and-play USB device—similar to the v2.0 in supporting USB Audio Class 1.0 for full compatibility with Windows, macOS, and Linux—yet tuned for out-of-the-box voice interface performance without requiring additional hardware.

Feature-wise, both support far-field voice capture and speech-enhancing algorithms like AEC (Acoustic Echo Cancellation), VAD (Voice Activity Detection), DOA (Direction of Arrival), Beamforming, and Noise Suppression, but the v3.0's algorithm optimizations deliver cleaner audio in real-world noisy environments.

The LED system remains at 12 programmable RGB LEDs in both versions, but the v3.0 is modeled after the newer USB 4 Mic Array design, making it smaller and simpler than the v2.0's developer-oriented form factor, while still retaining key professional voice interface features.

Version

Product Version	Changes	Released Date
ReSpeaker Mic Array v1.0	Initial	Aug 15, 2016
ReSpeaker Mic Array v2.0	XVSM-2000 is EOL,change MCU to XVF-3000 and reduce the Mics from 7 to 4.	Jan 25, 2018

Product Version	Changes	Released Date
ReSpeaker Mic Array v3.0	Codec Changed to TLV320AIC3104	Jan 19, 2021

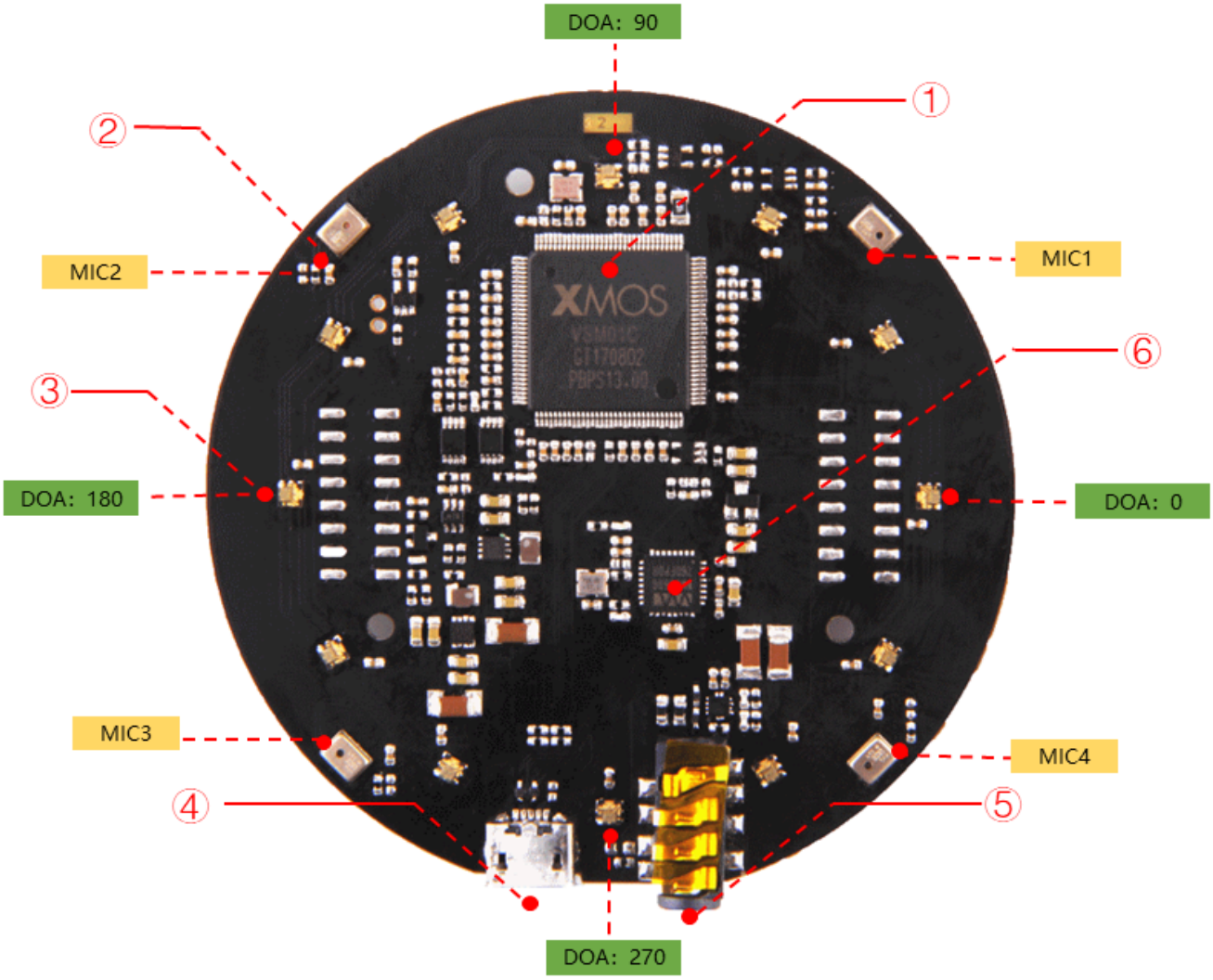
Features

- Far-field voice capture
- Support USB Audio Class 1.0 (UAC 1.0)
- Four microphones array
- 12 programmable RGB LED indicators
- Speech algorithms and features
  - Voice Activity Detection
  - Direction of Arrival
  - Beamforming
  - Noise Suppression
  - De-reverberation
  - Acoustic Echo Cancellation

Specification

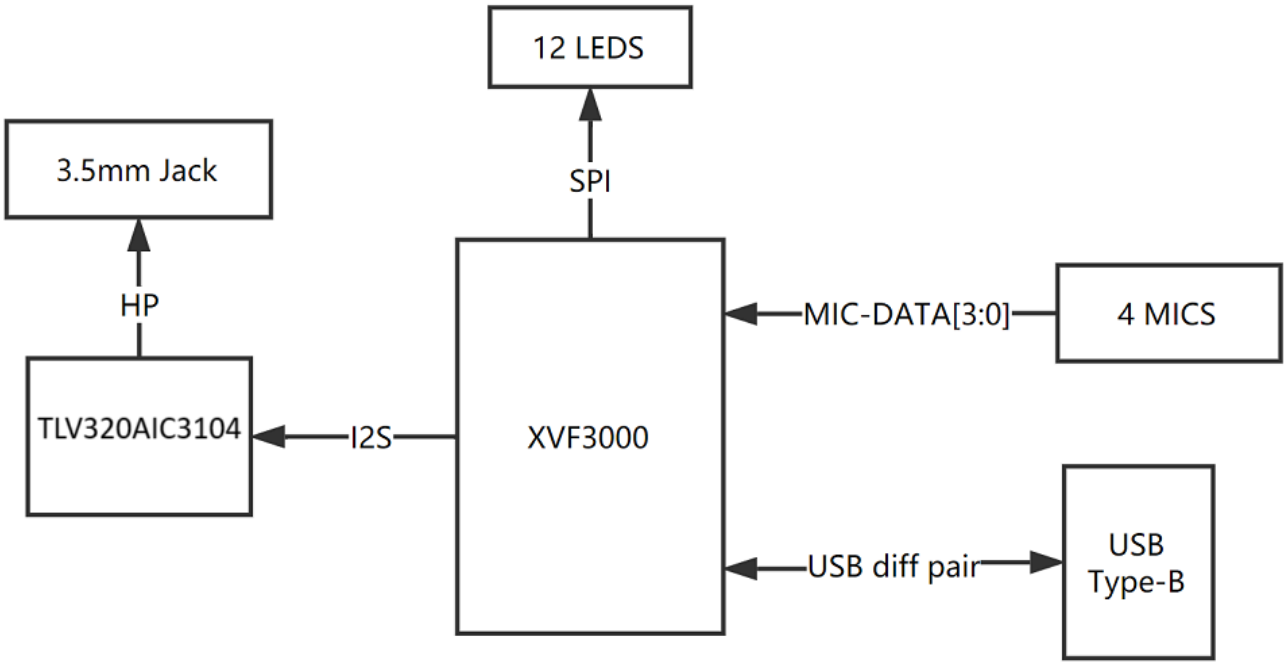
- XVF-3000 from XMOS
- 4 high performance digital microphones
- Supports Far-field Voice Capture
- Speech algorithm on-chip
- 12 programmable RGB LED indicators
- Microphones: ST MP34DT01TR-M
- Sensitivity: -26 dBFS (Omnidirectional)
- Acoustic overload point: 120 dBSPL
- SNR: 61 dB
- Power Supply: 5V DC from Micro USB or expansion header
- Dimensions: 70mm (Diameter)
- 3.5mm Audio jack output socket
- Power consumption: 5V, 180mA with led on and 170mA with led off
- Max Sample Rate:16Khz

Hardware Overview

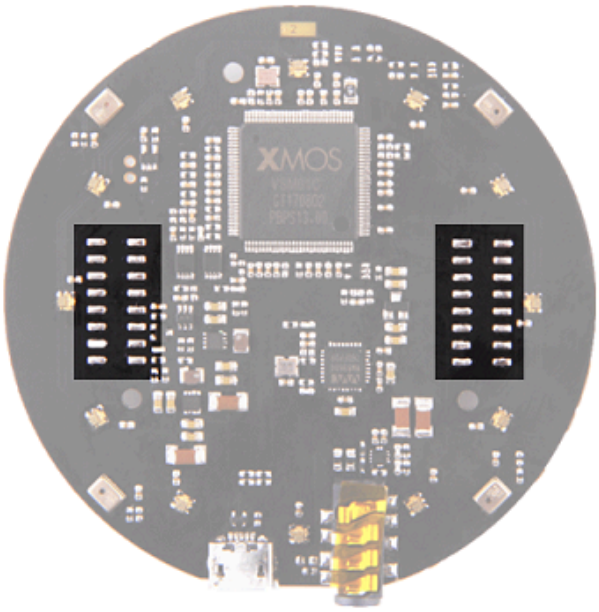


- ① **XMOS XVF-3000:** It integrates advanced DSP algorithms that include Acoustic Echo Cancellation (AEC), beamforming, dereverberation, noise suppression and gain control.
- ② **Digital Microphone:** The MP34DT01-M is an ultra-compact, lowpower, omnidirectional, digital MEMS microphone built with a capacitive sensing element and an IC interface.
- ③ **RGB LED:** Three-color RGB LED.
- ④ **USB Port:** Provide the power and control the mic array.
- ⑤ **3.5mm Headphone jack:** Output audio, We can plug active speakers or Headphones into this port.
- ⑥ **TLV320AIC3104:** The TLV320AIC3104 is a low power stereo codec featuring Class D speaker drivers to provide 1 W per channel into 8 W loads.

System Diagram



Pin Map



J5

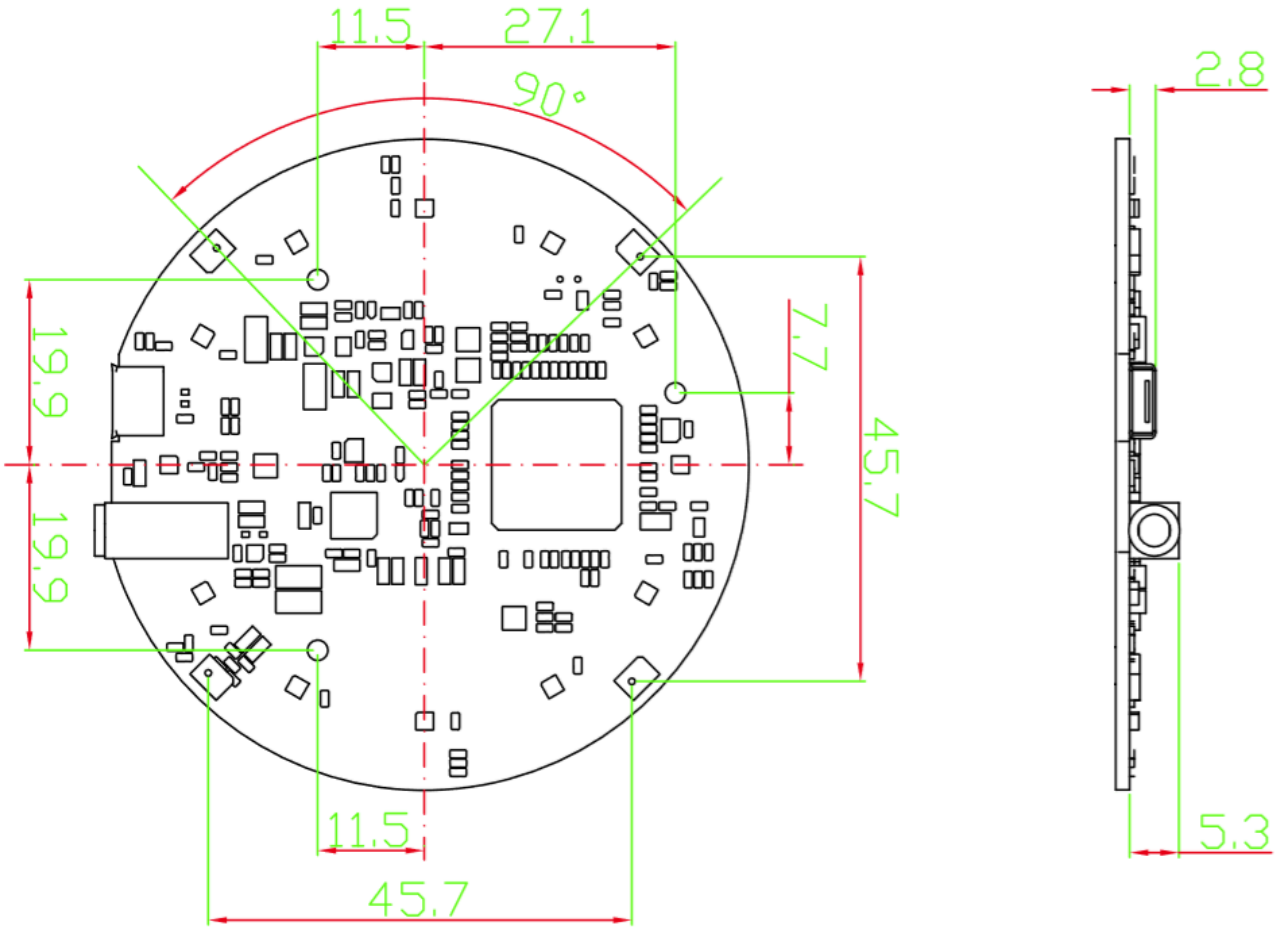
NC	NC
NC	NC
NC	NC
NC	NC
NC	NC
NC	NC
NC	USBDP
NC	USBDM

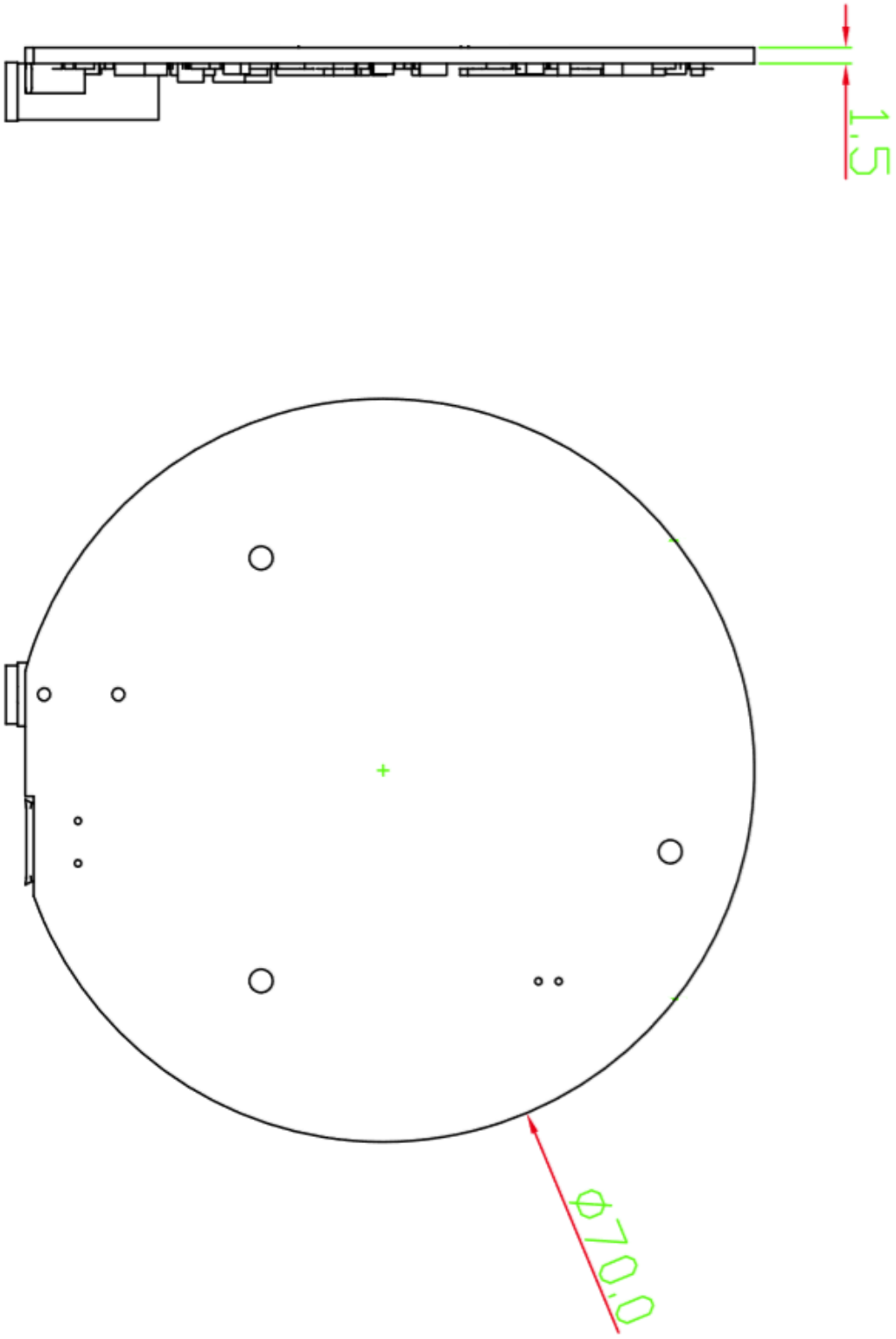
J6

VDD_5V	AUX_SEL
Central_Led	NC
NC	NC
HP_L	HP_R
NC	NC
NC	NC
GND	NC
NC	GND

GND
POWER
AUX_SEL/LED
SPEAKER
NC
USB

Dimensions




1.5
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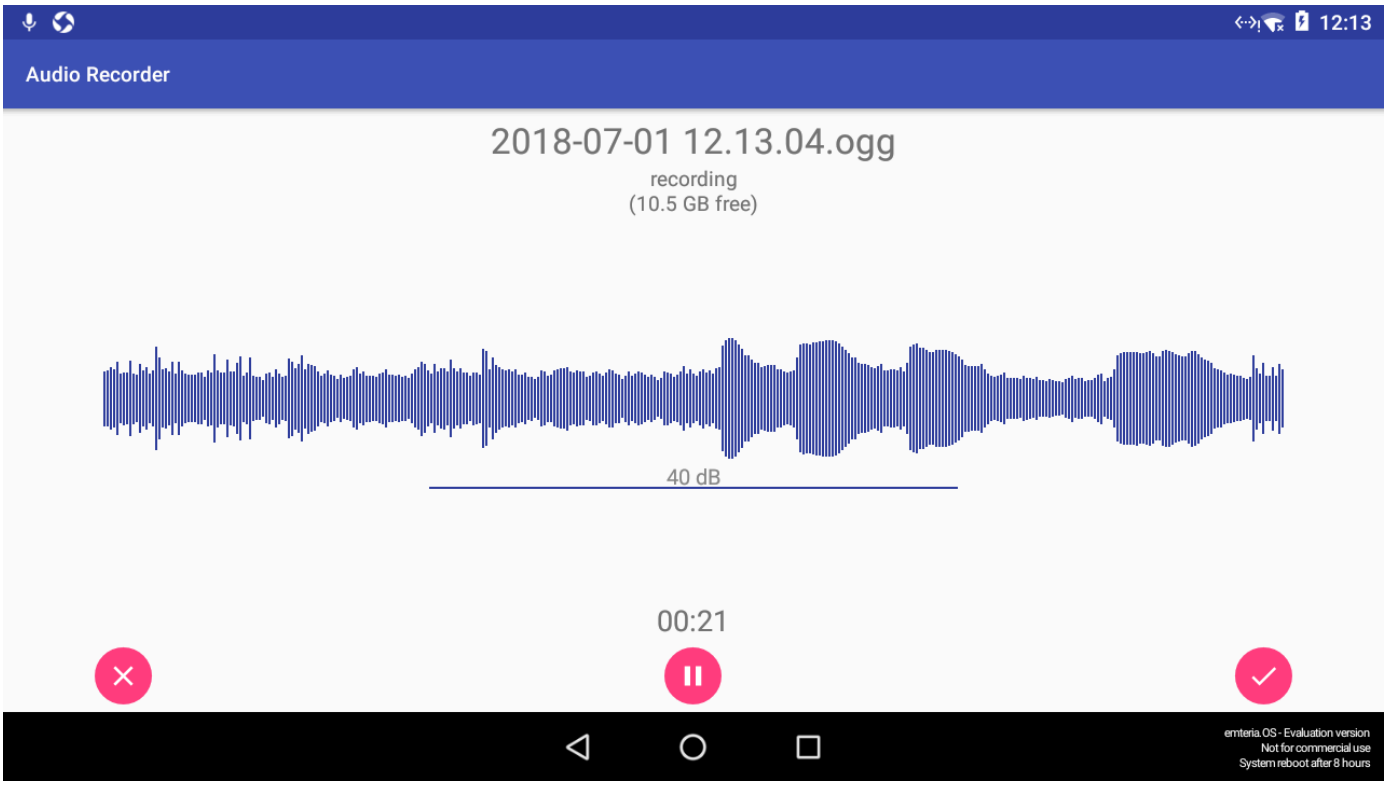
Applications

- USB Voice Capture
- Smart Speaker
- Intelligent Voice Assistant Systems
- Voice Recorders
- Voice Conferencing System
- Meeting Communicating Equipment
- Voice Interacting Robot
- Car Voice Assistant
- Other Voice Interface Scenarios

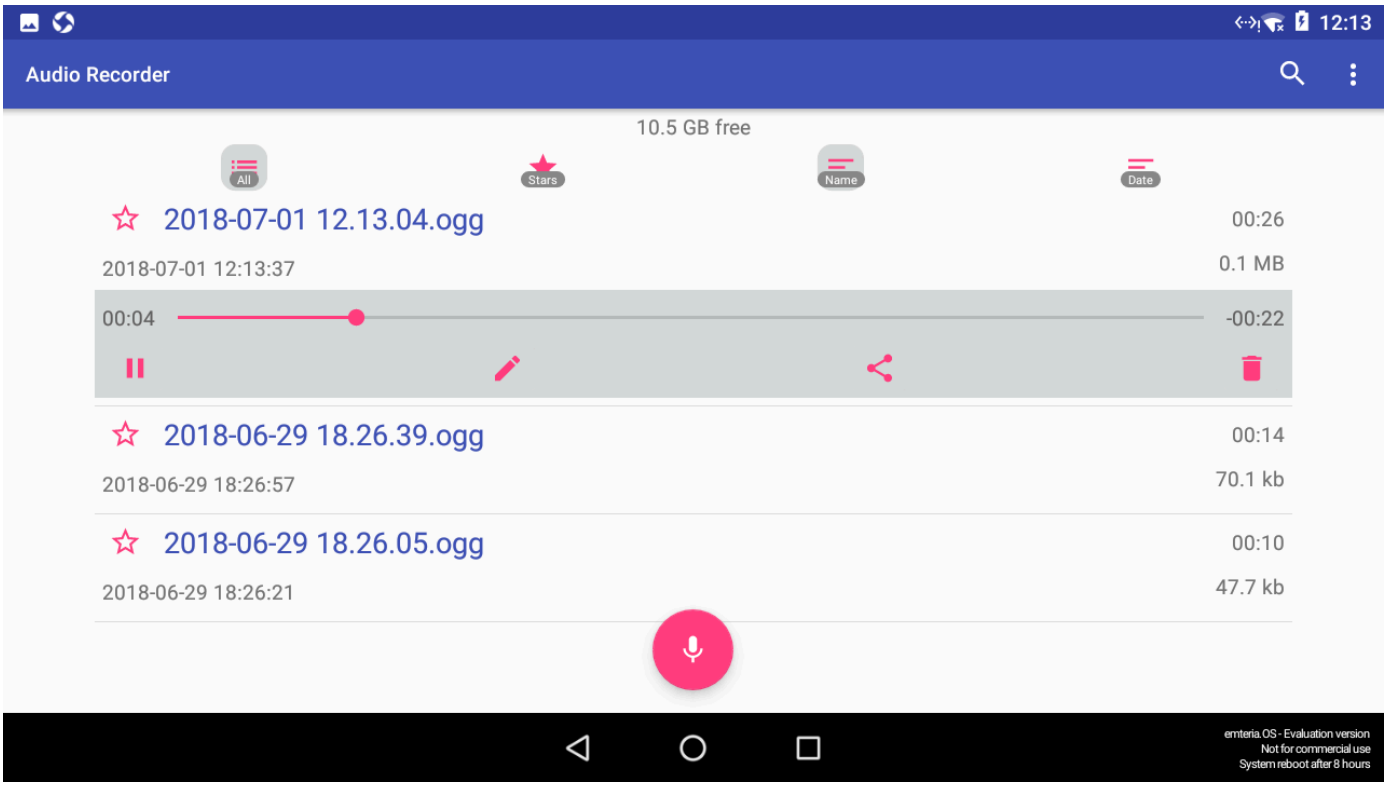
Getting Started

> **NOTE**
>
> ReSpeaker Mic Array v3.0 is compatiable with Windows, Mac, Linux systems andriod. The below scripts are tested on Python2.7.

For andriod, we tested it with emteria.OS(andriod 7.1) on Raspberry. We plug the mic array v3.0 to raspberry pi USB port and select the ReSpeaker mic array v3.0 as audio device. Here is the audio recording screen.



Here is the audio playing screen. We plug speaker to ReSpeaker mic array v3.0 3.5mm audio jack and hear what we record.



Update Firmware

There are 2 firmwares. One includes 1 channel data, while the other inlcudes 6 channels data (factory firmware). Here is the table for the differences.

Firmware	Channels	Note
1_channel_firmware.bin	1	Processed audio for ASR
6_channels_firmware.bin	6	Channel 0: processed audio for ASR Channel 1: mic1 raw data Channel 2: mic2 raw data Channel 3: mic3 raw data Channel 4: mic4 raw data Channel 5: merged playback

For Linux: The Mic array supports the USB DFU. We develop a python script dfu.py to update the firmware through USB.

```
sudo apt-get update
sudo pip install pyusb click
git clone https://github.com/respeaker/usb_4_mic_array.git
cd usb_4_mic_array
sudo python dfu.py --download MicArrayV3_firmware/6_channels_dfu_4.0.0_firmware.bin # The 6 channels version

# if you want to use 1 channel,then the command should be like:

sudo python dfu.py --download MicArrayV3_firmware/1_channel_dfu_4.0.0_firmware.bin
```

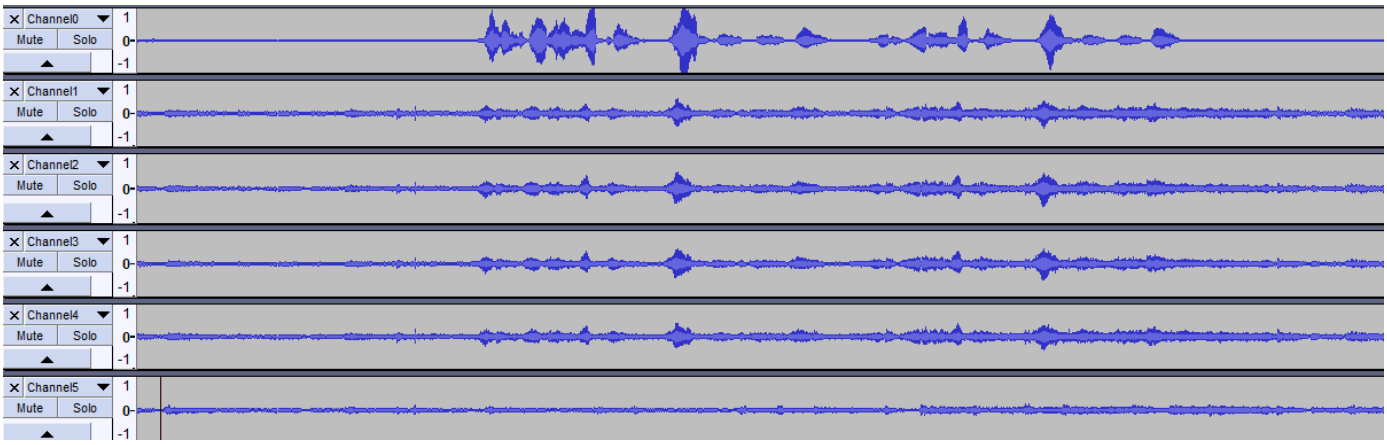
For Windows/Mac: We do not suggest use Windows/Mac and Linux vitual machine to update the firmware.

Out of Box Demo

Here is the Acoustic Echo Cancellation example with 6 channels firmware.

- Step 1. Connect the USB cable to PC and audio jack to speaker.

- Step 2. Select the mic array v3.0 as output device in PC side.
- Step 3. Start the audacity to record.
- Step 4. Play music at PC side first and then we talk.
- Step 5. We will see the audacity screen as below, Please click **Solo** to hear each channel audio.



Channel0 Audio(processed by algorithms):

0:00 / 0:12

Channel1 Audio(Mic1 raw data):

0:00 / 0:12

Channel5 Audio(Playback data):

0:00 / 0:12

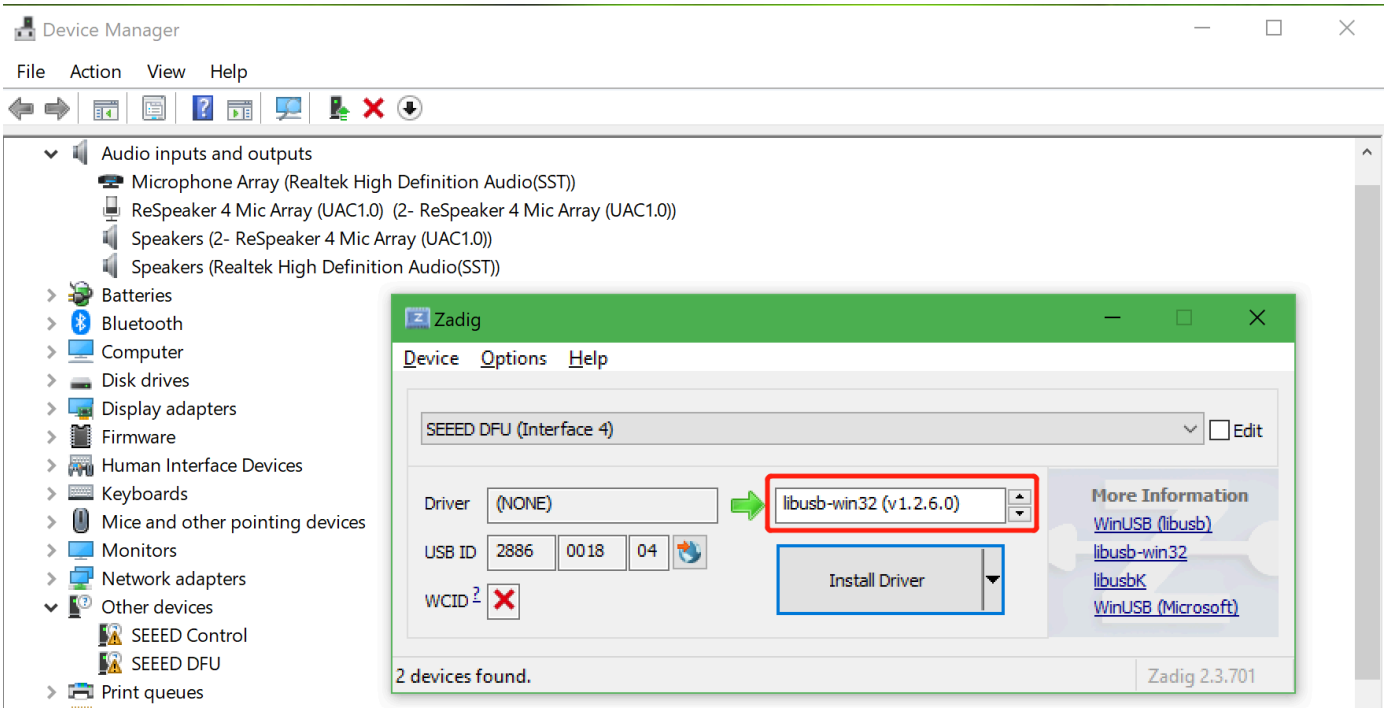
Here is the video about the DOA and AEC.

ReSpeaker Mic Array v2.0 - DOA & AEC Test
Seeed Studio

다음에서 보기:

Install DFU and LED Control Driver

- **Windows:** Audio recording and playback works well by default. Libusb-win32 driver is only required to control LEDs an DSP parameters on Windows. We use a handy tool - Zadig to install the libusb-win32 driver for both `SEEED DFU` and `SEEED Control` (ReSpeaker Mic Array has 2 devices on Windows Device Manager).



> **CAUTION**
>
> Please make sure that libusb-win32 is selected, not WinUSB or libusbK.

- **MAC:** No driver is required.
- **Linux:** No driver is required.

Tuning

For Linux/Mac/Windows: We can configure some parameters of built-in algorithms.

- Get the full list parameters, for more info, please refer to FAQ.

```
git clone https://github.com/respeaker/usb_4_mic_array.git
cd usb_4_mic_array
python tuning.py -p
```

- Example#1, we can turn off Automatic Gain Control (AGC):

```
python tuning.py AGCONOFF 0
```

- Example#2, We can check the DOA angle.

```
pi@raspberrypi:~/usb_4_mic_array $ sudo python tuning.py DOAANGLE
DOAANGLE: 180
```

Control the LEDs

We can control the ReSpeaker Mic Array V2's LEDs through USB. The USB device has a Vendor Specific Class Interface which can be used to send data through USB Control Transfer. We refer pyusb python library and come out the usb_pixel_ring python library.

The LED control command is sent by pyusb's usb.core.Device.ctrl_transfer(), its parameters as below :

```
ctrl_transfer(usb.util.CTRL_OUT | usb.util.CTRL_TYPE_VENDOR |
usb.util.CTRL_RECIPIENT_DEVICE, 0, command, 0x1C, data, TIMEOUT)
```

Here are the usb_pixel_ring APIs.

Command	Data	API	Note
0	[0]	pixel_ring.trace()	trace mode, LEDs changing depends on VAD*and DOA*
1	[red, green, blue, 0]	pixel_ring.mono()	mono mode, set all RGB LED to a single color, for example Red(0xFF0000), Green(0x00FF00) , Blue(0x0000FF)
2	[0]	pixel_ring.listen()	listen mode, similar with trace mode, but not turn LEDs off
3	[0]	pixel_ring.speak()	wait mode
4	[0]	pixel_ring.think()	speak mode
5	[0]	pixel_ring.spin()	spin mode
6	[r, g, b, 0] * 12	pixel_ring.custimize()	custom mode, set each LED to its own color
0x20	[brightness]	pixel_ring.set_brightness()	set brightness, range: 0x00~0x1F
0x21	[r1, g1, b1, 0, r2, g2, b2, 0]	pixel_ring.set_color_palette()	set color palette, for example, pixel_ring.set_color_palette(0xff0000, 0x00ff00) together with pixel_ring.think()
0x22	[vad_led]	pixel_ring.set_vad_led()	set center LED: 0 - off, 1 - on, else - depends on VAD
0x23	[volume]	pixel_ring.set_volume()	show volume, range: 0 ~ 12

Command	Data	API	Note
0x24	[pattern]	pixel_ring.change_pattern()	set pattern, 0 - Google Home pattern, others - Echo pattern

For Linux: Here is the example to control the leds. Please follow below commands to run the demo.

```
git clone https://github.com/respeaker/pixel_ring.git
cd pixel_ring
sudo python setup.py install
sudo python examples/usb_mic_array.py
```

Here is the code of the usb_mic_array.py.

```
import time
from pixel_ring import pixel_ring


if __name__ == '__main__':
    pixel_ring.change_pattern('echo')
    while True:

        try:
            pixel_ring.wakeup()
            time.sleep(3)
            pixel_ring.think()
            time.sleep(3)
            pixel_ring.speak()
            time.sleep(6)
            pixel_ring.off()
            time.sleep(3)
        except KeyboardInterrupt:
            break


    pixel_ring.off()
    time.sleep(1)
```

For Windows/Mac: Here is the example to control the leds.

- Step 1. Download pixel_ring.

```
git clone https://github.com/respeaker/pixel_ring.git
cd pixel_ring/pixel_ring
```

- Step 2. Create a led_control.py with below code and run 'python led_control.py'

```
from usb_pixel_ring_v2 import PixelRing
import usb.core
import usb.util
import time

dev = usb.core.find(idVendor=0x2886, idProduct=0x0018)
print dev
if dev:
    pixel_ring = PixelRing(dev)

    while True:
        try:
            pixel_ring.wakeup(180)
            time.sleep(3)
            pixel_ring.listen()
            time.sleep(3)
            pixel_ring.think()
            time.sleep(3)
            pixel_ring.set_volume(8)
            time.sleep(3)
            pixel_ring.off()
            time.sleep(3)
        except KeyboardInterrupt:
            break

    pixel_ring.off()
```

> **NOTE**
>
> If you see "None" printed on screen, please reinstall the libusb-win32 driver.

DOA (Direction of Arrival)

For Windows/Mac/Linux: Here is the example to view the DOA. The Green LED is the indicator of the voice direction. For the angle, please refer to hardware overview.

- Step 1. Download the usb_4_mic_array.

```
git clone https://github.com/respeaker/usb_4_mic_array.git
cd usb_4_mic_array
```

- Step 2. Create a DOA.py with below code under usb_4_mic_array folder and run 'python DOA.py'

> **NOTE**
>
> If you are using Python 3, this file is suitable for use DOA.py. And the tuning file is tuning.py

```
from tuning import Tuning
import usb.core
import usb.util
import time

dev = usb.core.find(idVendor=0x2886, idProduct=0x0018)

if dev:
    Mic_tuning = Tuning(dev)
    print Mic_tuning.direction
    while True:
        try:
            print Mic_tuning.direction
            time.sleep(1)
        except KeyboardInterrupt:
            break
```

- Step 3. We will see the DOA as below.

```
pi@raspberrypi:~/usb_4_mic_array $ sudo python doa.py
184
183
175
105
104
104
103
```

VAD (Voice Activity Detection)

For Windows/Mac/Linux: Here is the example to view the VAD. The Red LED is the indicator of the VAD.

- Step 1. Download the usb_4_mic_array.

```
git clone https://github.com/respeaker/usb_4_mic_array.git
cd usb_4_mic_array
```

- Step 2. Create a VAD.py with below code under usb_4_mic_array folder and run 'python VAD.py'

> **NOTE**
>
> If you are using Python 3, this file is suitable for use VAD.py.And the tuning file is tuning.py

```
from tuning import Tuning
import usb.core
import usb.util
import time
```

```
dev = usb.core.find(idVendor=0x2886, idProduct=0x0018)
#print dev
if dev:
    Mic_tuning = Tuning(dev)
    print Mic_tuning.is_voice()
    while True:
        try:
            print Mic_tuning.is_voice()
            time.sleep(1)
        except KeyboardInterrupt:
            break
```

- Step 3. We will see the DOA as below.

```
pi@raspberrypi:~/usb_4_mic_array $ sudo python VAD.py
0
0
0
1
0
1
0
```

> NOTE
>
> For the threshold of VAD, we also can use the GAMMAVAD_SR to set. Please refer to Tuning for more detail.

Extract Voice

We use PyAudio python library to extract voice through USB.

For Linux: We can use below commands to record or play the voice.

```
arecord -D plughw:1,0 -f cd test.wav # record, please use the arecord -l to check the card and hardware first
aplay -D plughw:1,0 -f cd test.wav # play, please use the aplay -l to check the card and hardware first
arecord -D plughw:1,0 -f cd |aplay -D plughw:1,0 -f cd # record and play at the same time
```

We also can use python script to extract voice.

- Step 1, We need to run the following script to get the device index number of Mic Array:

```
sudo pip install pyaudio
cd ~
```

```
nano get_index.py
```

- Step 2, copy below code and paste on get_index.py.

> **NOTE**
>
> If you are using Python 3, this file is suitable for use get_index.py

```
import pyaudio

p = pyaudio.PyAudio()
info = p.get_host_api_info_by_index(0)
numdevices = info.get('deviceCount')

for i in range(0, numdevices):
        if (p.get_device_info_by_host_api_device_index(0, i).get('maxInputChannels')) >
0:
            print "Input Device id ", i, " - ",
p.get_device_info_by_host_api_device_index(0, i).get('name')
```

- Step 3, press `Ctrl` + `X` to exit and press Y to save.
- Step 4, run 'sudo python get_index.py' and we will see the device ID as below.

```
Input Device id  2  -  ReSpeaker 4 Mic Array (UAC1.0): USB Audio (hw:1,0)
```

- Step 5, change `RESPEAKER_INDEX = 2` to index number. Run python script record.py to record a speech.

```
import pyaudio
import wave

RESPEAKER_RATE = 16000
RESPEAKER_CHANNELS = 6 # change base on firmwares, 1_channel_firmware.bin as 1 or
6_channels_firmware.bin as 6
RESPEAKER_WIDTH = 2
# run getDeviceInfo.py to get index
RESPEAKER_INDEX = 2  # refer to input device id
CHUNK = 1024
RECORD_SECONDS = 5
WAVE_OUTPUT_FILENAME = "output.wav"

p = pyaudio.PyAudio()

stream = p.open(
            rate=RESPEAKER_RATE,
            format=p.get_format_from_width(RESPEAKER_WIDTH),
            channels=RESPEAKER_CHANNELS,
```

```
            input=True,
            input_device_index=RESPEAKER_INDEX,)

print("* recording")

frames = []

for i in range(0, int(RESPEAKER_RATE / CHUNK * RECORD_SECONDS)):
    data = stream.read(CHUNK)
    frames.append(data)

print("* done recording")

stream.stop_stream()
stream.close()
p.terminate()

wf = wave.open(WAVE_OUTPUT_FILENAME, 'wb')
wf.setnchannels(RESPEAKER_CHANNELS)
wf.setsampwidth(p.get_sample_size(p.get_format_from_width(RESPEAKER_WIDTH)))
wf.setframerate(RESPEAKER_RATE)
wf.writeframes(b''.join(frames))
wf.close()
```

- Step 6. If you want to extract channel 0 data from 6 channels, please follow below code. For other channel X, please change [0::6] to [X::6].

```
import pyaudio
import wave
import numpy as np

RESPEAKER_RATE = 16000
RESPEAKER_CHANNELS = 6 # change base on firmwares, 1_channel_firmware.bin as 1 or
6_channels_firmware.bin as 6
RESPEAKER_WIDTH = 2
# run getDeviceInfo.py to get index
RESPEAKER_INDEX = 3  # refer to input device id
CHUNK = 1024
RECORD_SECONDS = 3
WAVE_OUTPUT_FILENAME = "output.wav"

p = pyaudio.PyAudio()

stream = p.open(
            rate=RESPEAKER_RATE,
            format=p.get_format_from_width(RESPEAKER_WIDTH),
            channels=RESPEAKER_CHANNELS,
            input=True,
            input_device_index=RESPEAKER_INDEX,)

print("* recording")
```

```
frames = []

for i in range(0, int(RESPEAKER_RATE / CHUNK * RECORD_SECONDS)):
    data = stream.read(CHUNK)
    # extract channel 0 data from 6 channels, if you want to extract channel 1, please
change to [1::6]
    a = np.fromstring(data,dtype=np.int16)[0::6]
    frames.append(a.tostring())

print("* done recording")

stream.stop_stream()
stream.close()
p.terminate()

wf = wave.open(WAVE_OUTPUT_FILENAME, 'wb')
wf.setnchannels(1)
wf.setsampwidth(p.get_sample_size(p.get_format_from_width(RESPEAKER_WIDTH)))
wf.setframerate(RESPEAKER_RATE)
wf.writeframes(b''.join(frames))
wf.close()
```

For Windows:

- Step 1. We run below command to install pyaudio.

```
pip install pyaudio
```

- Step 2. Use get_index.py to get device index.

```
C:\Users\XXX\Desktop>python get_index.py
Input Device id  0  -  Microsoft Sound Mapper - Input
Input Device id  1  -  ReSpeaker 4 Mic Array (UAC1.0)
Input Device id  2  -  Internal Microphone (Conexant I)
```

- Step 3. Modify the device index and channels of record.py and then extract voice.

```
C:\Users\XXX\Desktop>python record.py
* recording
* done recording
```

> ⚠ **CAUTION**
>
> If we see "Error: %1 is not a valid Win32 application.", please install Python Win32 version.

For MAC:

- Step 1. We run below command to install pyaudio.

```
pip install pyaudio
```

- Step 2. Use get_index.py to get device index.

```
MacBook-Air:Desktop XXX$ python get_index.py
Input Device id  0  -  Built-in Microphone
Input Device id  2  -  ReSpeaker 4 Mic Array (UAC1.0)
```

- Step 3. Modify the device index and channels of record.py and then extract voice.

```
MacBook-Air:Desktop XXX$ python record.py
2018-03-24 14:53:02.400 Python[2360:16629] 14:53:02.399 WARNING:  140: This application,
or a library it uses, is using the deprecated Carbon Component Manager for hosting Audio
Units. Support for this will be removed in a future release. Also, this makes the host
incompatible with version 3 audio units. Please transition to the API's in
AudioComponent.h.
* recording
* done recording
```

FAQ

Q1: Parameters of built-in algorithms

```
pi@raspberrypi:~/usb_4_mic_array $ python tuning.py -p
name   type max min r/w info
-------------------------------
AECFREEZEONOFF   int 1 0 rw Adaptive Echo Canceler updates inhibit.
                                                    0 = Adaptation enabled
                                                    1 = Freeze adaptation, filter
only
AECNORM          float 16 0.25 rw Limit on norm of AEC filter coefficients
AECPATHCHANGE    int 1 0 ro AEC Path Change Detection.
                                                    0 = false (no path change
detected)
                                                    1 = true (path change
detected)
AECSILENCELEVEL  float 1 1e-09 rw Threshold for signal detection in AEC [-inf .. 0] dBov
(Default: -80dBov = 10log10(1x10-8))
AECSILENCEMODE   int 1 0 ro AEC far-end silence detection status.
                                                    0 = false (signal detected)
                                                    1 = true (silence detected)
AGCDESIREDLEVEL  float 0.99 1e-08 rw Target power level of the output signal.
                                                    [-inf .. 0] dBov (default:
-23dBov = 10log10(0.005))
AGCGAIN          float 1000 1 rw Current AGC gain factor.
```

```
                                                        [0 .. 60] dB (default: 0.0dB
= 20log10(1.0))
AGCMAXGAIN       float 1000 1 rw Maximum AGC gain factor.
                                                        [0 .. 60] dB (default 30dB =
20log10(31.6))
AGCONOFF         int 1 0 rw Automatic Gain Control.
                                                        0 = OFF
                                                        1 = ON
AGCTIME          float 1 0.1 rw Ramps-up / down time-constant in seconds.
CNIONOFF         int 1 0 rw Comfort Noise Insertion.
                                                        0 = OFF
                                                        1 = ON
DOAANGLE         int 359 0 ro DOA angle. Current value. Orientation depends on build
configuration.
ECHOONOFF        int 1 0 rw Echo suppression.
                                                        0 = OFF
                                                        1 = ON
FREEZEONOFF      int 1 0 rw Adaptive beamformer updates.
                                                        0 = Adaptation enabled
                                                        1 = Freeze adaptation, filter
only
FSBPATHCHANGE    int 1 0 ro FSB Path Change Detection.
                                                        0 = false (no path change
detected)
                                                        1 = true (path change
detected)
FSBUPDATED       int 1 0 ro FSB Update Decision.
                                                        0 = false (FSB was not
updated)
                                                        1 = true (FSB was updated)
GAMMAVAD_SR      float 1000 0 rw Set the threshold for voice activity detection.
                                                        [-inf .. 60] dB (default:
3.5dB 20log10(1.5))
GAMMA_E          float 3 0 rw Over-subtraction factor of echo (direct and early
components). min .. max attenuation
GAMMA_ENL        float 5 0 rw Over-subtraction factor of non-linear echo. min .. max
attenuation
GAMMA_ETAIL      float 3 0 rw Over-subtraction factor of echo (tail components). min ..
max attenuation
GAMMA_NN         float 3 0 rw Over-subtraction factor of non- stationary noise. min ..
max attenuation
GAMMA_NN_SR      float 3 0 rw Over-subtraction factor of non-stationary noise for ASR.
                                                        [0.0 .. 3.0] (default: 1.1)
GAMMA_NS         float 3 0 rw Over-subtraction factor of stationary noise. min .. max
attenuation
GAMMA_NS_SR      float 3 0 rw Over-subtraction factor of stationary noise for ASR.
                                                        [0.0 .. 3.0] (default: 1.0)
HPFONOFF         int 3 0 rw High-pass Filter on microphone signals.
                                                        0 = OFF
                                                        1 = ON - 70 Hz cut-off
                                                        2 = ON - 125 Hz cut-off
                                                        3 = ON - 180 Hz cut-off
MIN_NN           float 1 0 rw Gain-floor for non-stationary noise suppression.
```

```
                                                        [-inf .. 0] dB (default:
-10dB = 20log10(0.3))
MIN_NN_SR        float 1 0 rw Gain-floor for non-stationary noise suppression for ASR.
                                                        [-inf .. 0] dB (default:
-10dB = 20log10(0.3))
MIN_NS           float 1 0 rw Gain-floor for stationary noise suppression.
                                                        [-inf .. 0] dB (default:
-16dB = 20log10(0.15))
MIN_NS_SR        float 1 0 rw Gain-floor for stationary noise suppression for ASR.
                                                        [-inf .. 0] dB (default:
-16dB = 20log10(0.15))
NLAEC_MODE       int 2 0 rw Non-Linear AEC training mode.
                                                        0 = OFF
                                                        1 = ON - phase 1
                                                        2 = ON - phase 2
NLATTENONOFF     int 1 0 rw Non-Linear echo attenuation.
                                                        0 = OFF
                                                        1 = ON
NONSTATNOISEONOFF int 1 0 rw Non-stationary noise suppression.
                                                        0 = OFF
                                                        1 = ON
NONSTATNOISEONOFF_SR int 1 0 rw Non-stationary noise suppression for ASR.
                                                        0 = OFF
                                                        1 = ON
RT60             float 0.9 0.25 ro Current RT60 estimate in seconds
RT60ONOFF        int 1 0 rw RT60 Estimation for AES. 0 = OFF 1 = ON
SPEECHDETECTED   int 1 0 ro Speech detection status.
                                                        0 = false (no speech
detected)
                                                        1 = true (speech detected)
STATNOISEONOFF   int 1 0 rw Stationary noise suppression.
                                                        0 = OFF
                                                        1 = ON
STATNOISEONOFF_SR int 1 0 rw Stationary noise suppression for ASR.
                                                        0 = OFF
                                                        1 = ON
TRANSIENTONOFF   int 1 0 rw Transient echo suppression.
                                                        0 = OFF
                                                        1 = ON
VOICEACTIVITY    int 1 0 ro VAD voice activity status.
                                                        0 = false (no voice activity)
                                                        1 = true (voice activity)
```

Q2: ImportError: No module named usb.core

A2: Run sudo pip install pyusb to install the pyusb.

```
pi@raspberrypi:~/usb_4_mic_array $ sudo python tuning.py DOAANGLE
Traceback (most recent call last):
  File "tuning.py", line 5, in <module>
    import usb.core
ImportError: No module named usb.core
```

```
pi@raspberrypi:~/usb_4_mic_array $ sudo pip install pyusb
Collecting pyusb
  Downloading pyusb-1.0.2.tar.gz (54kB)
    100% |████████████████████████████████| 61kB 101kB/s
Building wheels for collected packages: pyusb
  Running setup.py bdist_wheel for pyusb ... done
  Stored in directory:
/root/.cache/pip/wheels/8b/7f/fe/baf08bc0dac02ba17f3c9120f5dd1cf74aec4c54463bc85cf9
Successfully built pyusb
Installing collected packages: pyusb
Successfully installed pyusb-1.0.2
pi@raspberrypi:~/usb_4_mic_array $ sudo python tuning.py DOAANGLE
DOAANGLE: 180
```

Q3: Do you have the example for Raspberry alexa application?

A3: Yes, we can connect the mic array v3.0 to raspberry usb port and follow Raspberry Pi Quick Start Guide with Script to do the voice interaction with alexa.

Q4: Do you have the example for Mic array v3.0 with ROS system?

A4: Yes, thanks for Yuki sharing the package for integrating ReSpeaker Mic Array v2 with ROS (Robot Operating System) Middleware.

Q5: How to enable 3.5mm audio port to receive the signal as well as usb port?

A5: Please download the new firmware and burn the XMOS by following How to update firmware.

Resource

- **[PDF]** ReSpeaker MicArray v3.0 Schematic
- **[PDF]** ReSpeaker MicArray v3.0 Product Brief
- **[PDF]** ReSpeaker MicArray v3.0 3D Model
- **[SKP]** ReSpeaker MicArray v3.0 3D Model
- **[STP]** ReSpeaker MicArray v3.0 3D Model
- **[PDF]** XVF3000 Product Brief
- **[PDF]** XVF3000 Datasheet
- **[Github]** ReSpeaker Mic Array v2 with ROS (Robot Operating System) Middleware

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*Last updated on **Aug 13, 2025** by **Kasun Thushara***